

BRA-EUR - Towards Refining the Network Analysis

```
source(here::here("_chapter-setup.R")) # all standard packages and defaults
```

Overview

The idea is to develop a refined network analysis.

For this we load the data for 2024 for Brazil (i.e. ANAC data set) and Europe (NM flight table).

```
# ----- load the Brazilian network data -----
cols <- c(
  "sg_empresa_icao",
  "nr_voo",
  "nm_mes_partida_real",
  "nr_mes_partida_real",
  "sg_icao_origem",
  "sg_icao_destino",
  "lt_combustivel",
  "sg Equipamento_icao",
  "nm_regiao_origem",
  "nm_pais_origem",
  "nm_regiao_destino",
  "nm_pais_destino",
  "nr_horas_voadas",
  NULL
)

bra_mvts <- read_csv("./data_proc/flights_ANAC_2024.csv"
  , col_select = dplyr::any_of(cols)
```

```

        , show_col_types = FALSE
      ) |>
  rename(
    ADEP = sg_icao_origem,
    ADES = sg_icao_destino,
    OPR  = sg_empresa_icao,          # aircraft operator (ICAO)
    FUEL_BURN = lt_combustivel,
    TYPE = sg_equipamento_icao,     # aircraft type (ICAO)
    REGIAO_DEP = nm_regiao_origem,
    DEP_CNTRY = nm_pais_origem,     # departing country
    REGIAO_ARR = nm_regiao_destino,
    DES_CNTRY = nm_pais_destino,    # destination country
    FLT_DUR = nr_horas_voadas      # flight duration
  )

```

New names:

```
* `` -> `...1`
```

```

# ----- Europe -----
# data is stored on local network drive
source(here::here("R","rqutils-zip.R"))
pth_to_data <- here::here(here::here() |> dirname(), "__DATA","NM-flight-table")
payload_zip <- "NM-flt-2024.zip"

check_zip_content(pth_to_data, payload_zip) # check what we have in zip

```

	Name	Length	Date
1	NM-FLT-2024-Q1.gz.parquet	166156360	2024-04-17 13:27:00
2	NM-FLT-2024-Q2.gz.parquet	226135599	2024-07-17 11:39:00
3	NM-FLT-2024-Q3.gz.parquet	228081776	2024-10-03 12:54:00
4	NM-FLT-2024-Q4.gz.parquet	197557076	2025-03-31 09:48:00

```

eur_mvts <- read_zip(pth_to_data, payload_zip)
apts_meta <- arrow::read_parquet("./data/airport-icao-meta.parquet")

#| Utility function to prepare data -----
#|
prep_eur_network_data <- function(.eur_mvts, .apts_meta = apts_meta){
  this_mvts <- .eur_mvts |>
    dplyr::select(ADEP, ADES, FLTID = AIRCRAFT_ID, OPR = AIRCRAFT_OPERATOR
      , REG = REGISTRATION, TYPE = AIRCRAFT_TYPE_ICAO_ID

```

```

        , FLT_DUR = FLT_DUR_3
        , DATE = LOBT
        , SAM_ID = ID
      ) |>
    dplyr::mutate(DATE = date(DATE))

# augment airport details
this_mvts <- this_mvts |>
  dplyr::left_join(.apts_meta |>
    dplyr::select(ADEP = ICAO, DEP_CNTRY = COUNTRY_ISO)
    , by = dplyr::join_by(ADEP)) |>
  dplyr::left_join(.apts_meta |>
    dplyr::select(ADES = ICAO, DES_CNTRY = COUNTRY_ISO)
    , by = dplyr::join_by(ADES))

  return(this_mvts)
}

eur_mvts <- eur_mvts |> dplyr::bind_rows() |> prep_eur_network_data()
# all 2024 flights within EUR airspace -----
# eur_mvts |> arrow::write_parquet(
#   "./data-prep-eur/EUR-network-connectivity-analytic-2024.parquet"
#)
# write out study airport data -----
# eur_mvts |>
#   filter(ADEP %in% eur_aps | ADES %in% eur_aps) |>
#   arrow::write_parquet(
#     "./data/EUR-network-connectivity-analytic-2024.parquet")

calc_summary_routes_dep <- function(.mvts){
  this_summary <- .mvts |>
    dplyr::group_by(ADEP) |>
    dplyr::reframe(
      DEPS = dplyr::n()
      ,NBR_DES = dplyr::n_distinct(ADES)) |>
    dplyr::arrange(dplyr::desc(NBR_DES))

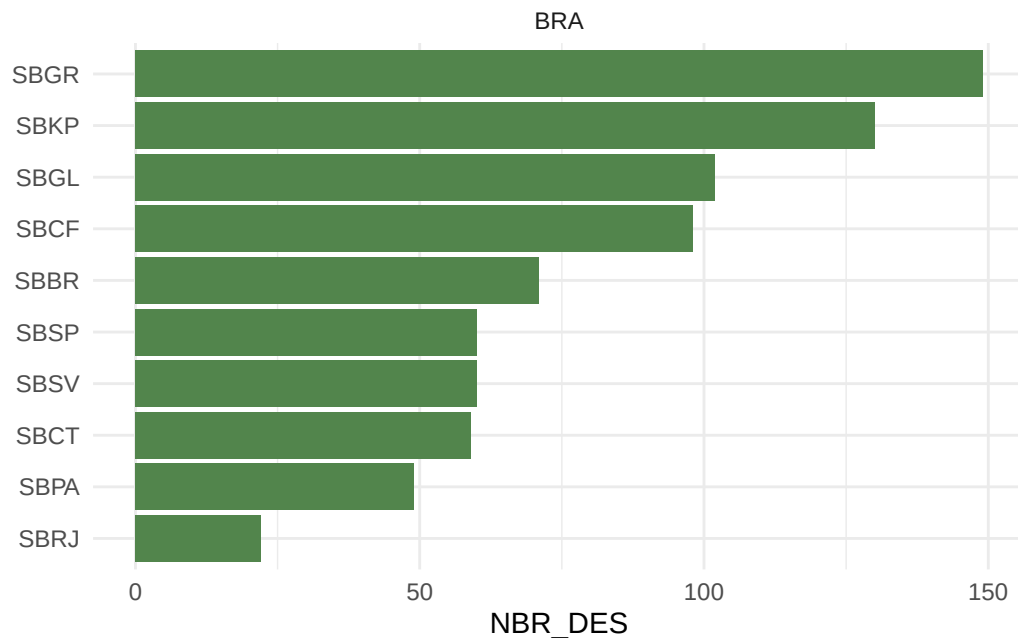
  return(this_summary)
}

bra_deps_routes <- bra_mvts |> calc_summary_routes_dep()
eur_deps_routes <- eur_mvts |> bind_rows() |> calc_summary_routes_dep()

```

```
# TODO - save as interim data and work from there
```

```
bind_rows(  
  bra_deps_routes |> filter(ADEP %in% bra_apt) |> mutate(REGION = "BRA")  
  # ,eur_deps_routes |> filter(ADEP %in% eur_apt) |> mutate(REGION = "EUR")  
  ) |>  
  mutate(ADEP = fct_infreq(ADEP, NBR_DES) |> fct_rev()) |>  
  ggplot() +  
  geom_col(aes(x = NBR_DES, y = ADEP, fill = REGION)) +  
  scale_fill_manual(values = bra_eur_colours) +  
  facet_wrap(~ REGION, scales = "free_y") +  
  labs(y = element_blank()) +  
  theme(legend.position = "none")
```



```
get_spread_of_destinations <- function(.deps){  
  this_spread <- .deps |>  
    dplyr::group_by(ADEP, DES_CNTRY) |>  
    dplyr::reframe(  
      DEPS = dplyr::n()  
      , DEST = dplyr::n_distinct(ADES)) |>  
    dplyr::group_by(ADEP) |>
```

```

    dplyr::mutate(
      SHARE_DEP      = DEPS / sum(DEPS)
      , TOP_DEST_SHARE = DEST / sum(DEST)
    ) |>
    dplyr::arrange(desc(DEPS), .by_group = TRUE) |>
    dplyr::ungroup()

    return(this_spread)
  }

#----- BRA -----
bra_dests <- bra_mvts |>
  filter(ADEP %in% bra_aps) |>
  get_spread_of_destinations()

#----- EUR -----
eur_mvts_study <- arrow::read_parquet("./data/EUR-network-connectivity-analytic-2024.parquet")
ectl_ms        <- readxl::read_xlsx("./data/ECTRL-ICAOEUR-state-memberships.xlsx") |> filter(
  mutate(REGION = "EUR")
)

#----- STATE and ISO CODES -----
global_iso2    <- ISOcodes::ISO_3166_1 |>
  tibble() |>
  select(STATE_ISO = Alpha_2, STATE = Name)

patch_global <- tibble::tribble(
  ~ STATE_ISO, ~STATE,
  "AN", "Netherlands Antilles",
  "XK", "Kosovo"           # need to think how to add to EUR region
)
global_iso2 <- global_iso2 |> bind_rows(patch_global)

# world wide codes - we may need this for generalising BRA data
# constructed wiht ChatGPT - not validated !!!!
icao_iso2 <- read_csv("./data/country-icao-iso-territory.csv")

```

Rows: 218 Columns: 4

-- Column specification -----

Delimiter: ","

chr (4): ICAO Region, Country/Territory Name, ISO Alpha-2 Code, Sovereign State

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

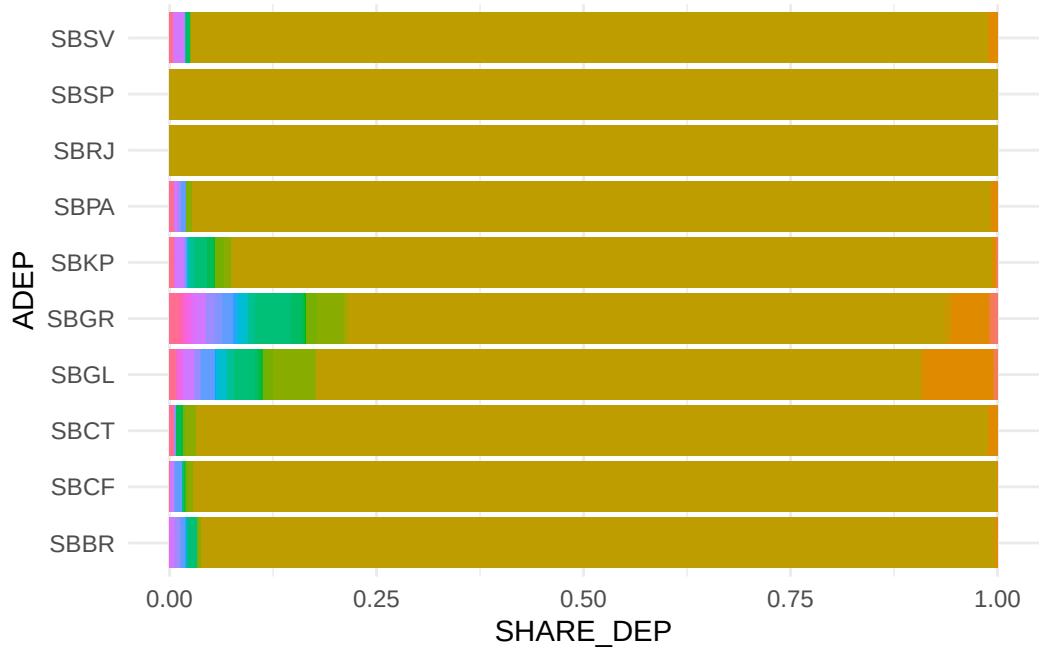
```
#----- END ISO -----

eur_dests <- eur_mvts_study |>
  filter(ADEP %in% eur_aps) |>
  get_spread_of_destinations() |>
  #----- append ECTRL / EUR -----
  left_join(global_iso2 |>
    rename(DES_CNTRY = STATE_ISO), by = join_by(DES_CNTRY)
  ) |>
  left_join(ectrl_ms |> select(-STATE) |>
    rename(DES_CNTRY = STATE_ISO) , by = join_by(DES_CNTRY)
  )
```

```
p <- bra_dests |>
  ggplot() + geom_col(aes(x = SHARE_DEP, y = ADEP, fill = DES_CNTRY)) +
  theme(
    legend.text = element_text(size = 7),
    legend.key.size = unit(0.3, "lines")
  )
p
```



```
p + theme(legend.position = "none")
```



international traffic

```
bra_dests |> filter(DES_CNTRY != "BRASIL") |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev())
```

A tibble: 145 x 6

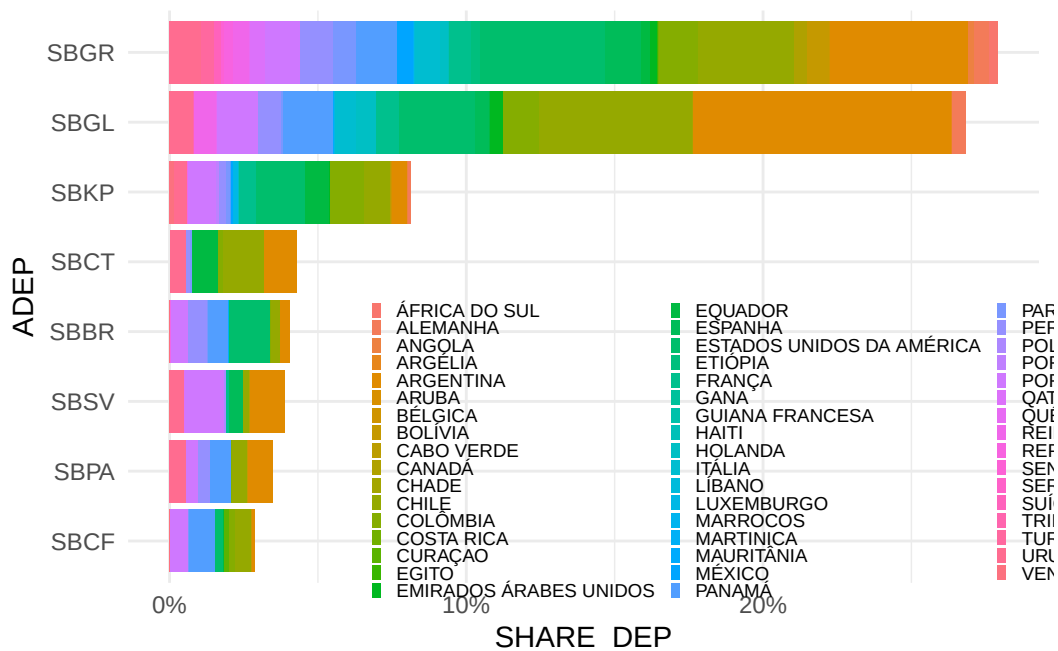
	ADEP	DES_CNTRY	DEPS	DEST	SHARE_DEP	TOP_DEST_SHARE
	<fct>	<chr>	<int>	<int>	<dbl>	<dbl>
1	SBBR	ESTADOS UNIDOS DA AMÉRICA	770	3	0.0141	0.0423
2	SBBR	PANAMÁ	372	1	0.00682	0.0141
3	SBBR	PERU	365	1	0.00670	0.0141
4	SBBR	PORTUGAL	326	1	0.00598	0.0141
5	SBBR	ARGENTINA	194	2	0.00356	0.0282
6	SBBR	CHILE	165	2	0.00303	0.0282
7	SBBR	URUGUAI	8	1	0.000147	0.0141
8	SBBR	MÉXICO	7	1	0.000128	0.0141
9	SBBR	VENEZUELA	3	1	0.0000550	0.0141
10	SBBR	MARROCOS	2	1	0.0000367	0.0141

i 135 more rows

```
# DES_CNTRY ~ PAIS_DESTINO

bra_dests |> filter(DES_CNTRY != "BRASIL") |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev()) |>

ggplot() + geom_col(aes(x = SHARE_DEP, y = ADEP, fill = DES_CNTRY)) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(fill = element_blank()) +
  theme(
    legend.text = element_text(size = 7),
    legend.key.size = unit(0.3, "lines"),
    legend.position = "inside", legend.position.inside = c(0.75,0.25)
  )
```



```
eur_dests |>
  filter(is.na(REGION)) |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev()) |>

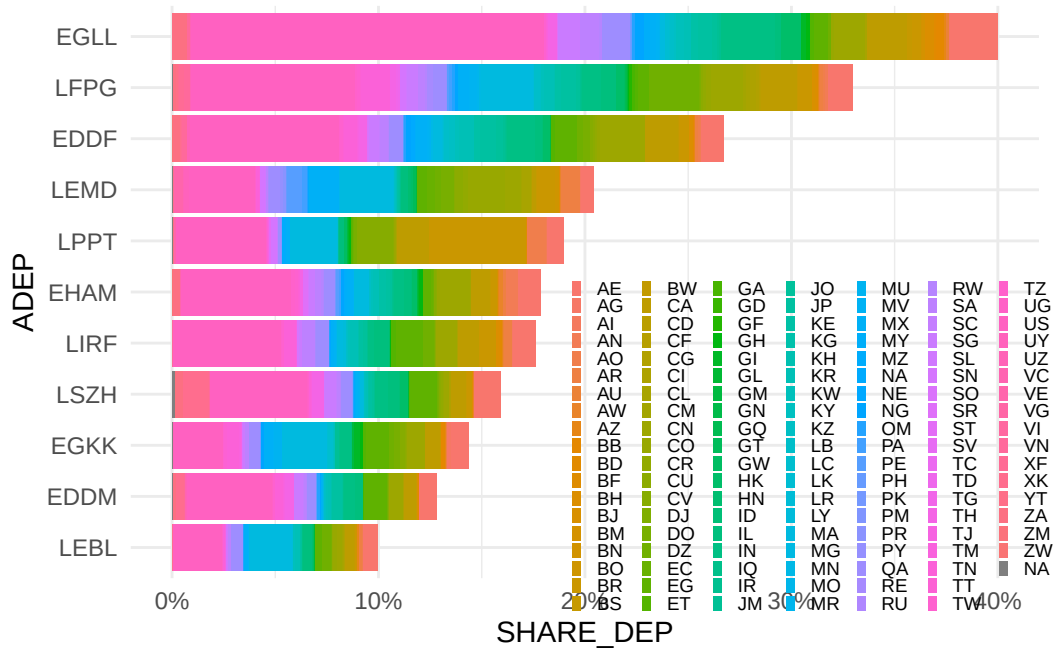
ggplot() + geom_col(aes(x = SHARE_DEP, y = ADEP, fill = DES_CNTRY)) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(fill = element_blank()) +
  theme()
```



```

legend.text = element_text(size = 7),
legend.key.size = unit(0.3, "lines"),
legend.position = "inside", legend.position.inside = c(0.75,0.25)
)

```



Appendix 1 to the Directives to Regional Air Navigation Meetings and Rules of Procedure for their Conduct (Doc 8144-AN/874), comprises the following regions:

- Africa-Indian Ocean Region (AFI);
- Asia Region (ASIA);
- Caribbean Region (CAR);
- European Region (EUR);
- Middle East Region (MID);
- North American Region (NAM);
- North Atlantic Region (NAT);
- Pacific Region (PAC);
- South American Region (SAM).

```

eur_world_shares <- eur_dests |>
  filter(is.na(REGION)) |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev()) |>
  left_join(icao_iso2 |> rename(DES_CNTRY = `ISO Alpha-2 Code`, ICAO_REG = `ICAO Region`) |>
    select(DES_CNTRY, ICAO_REG)) |>

```

```
# aggregate by ICAO Region
group_by(ICA0_REG) |>
reframe(DEPS = sum(DEPS), DEST = sum(DEST)) |>
mutate(REG_SHARES = DEPS / sum(DEPS))
```

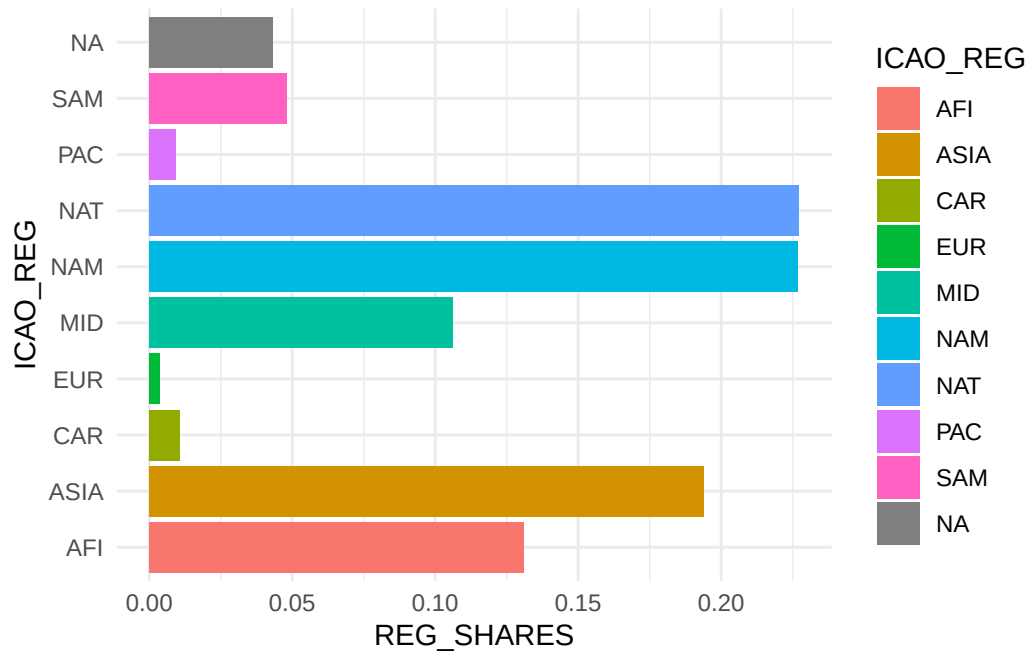
Joining with `by = join\_by(DES\_CNTRY)`

Warning in left_join(mutate(filter(eur_dests, is.na(REGION)), ADEP = fct_rev(fct_infreq(ADEP
i Row 1 of `x` matches multiple rows in `y`.
i Row 175 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
"many-to-many"` to silence this warning.

eur_world_shares

```
# A tibble: 10 x 4
  ICAO_REG  DEPS  DEST REG_SHARES
  <chr>    <int> <int>    <dbl>
1 AFI      83542   705    0.131
2 ASIA    123706   557    0.194
3 CAR       6763    80    0.0106
4 EUR       2426    9    0.00380
5 MID     67792   277    0.106
6 NAM    144737   675    0.227
7 NAT    145065   683    0.227
8 PAC       5861   13    0.00918
9 SAM     30728   114    0.0482
10 <NA>    27539   139    0.0432
```

```
eur_world_shares |>
ggplot() +
geom_col(aes(x = REG_SHARES, y = ICAO_REG, fill = ICAO_REG))
```



EUR ~ non-ECTRL Europe (probably need to take this out)

```
calc_adeq_counts <- function(.ds){
  my_count <- .ds |>
  group_by(ADEP, DEP_CNTRY) |> # Group by ADEP and PAIS_ORIGEM
  summarise(count = n(), .groups = "drop") |>
  arrange(desc(count))

  return(my_count)
}
```

```
bra_mvts |> calc_adeq_counts()
```

```
# A tibble: 361 x 3
  ADEP DEP_CNTRY count
<chr> <chr>    <int>
1 SBGR BRASIL 136561
2 SBSP BRASIL  94262
3 SBKP BRASIL  59915
4 SBCF BRASIL  55898
5 SBBR BRASIL  54509
```

```
6 SBGL  BRASIL    48774
7 SBRF  BRASIL    41092
8 SBRJ  BRASIL    28620
9 SBSV  BRASIL    27645
10 SBCT  BRASIL    26098
# i 351 more rows
```

```
plot_barchart <- function(.ds){

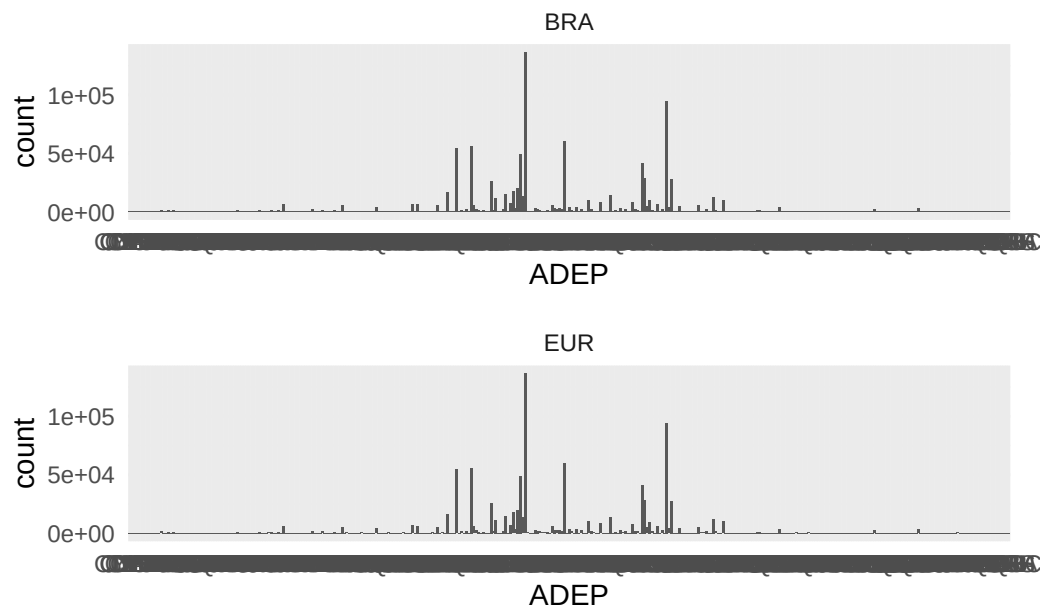
  p <- .ds |>
  ggplot() +
  geom_col(aes(x = ADEP, y = count)) +
    facet_wrap(~ REGION)

  return(p)
}

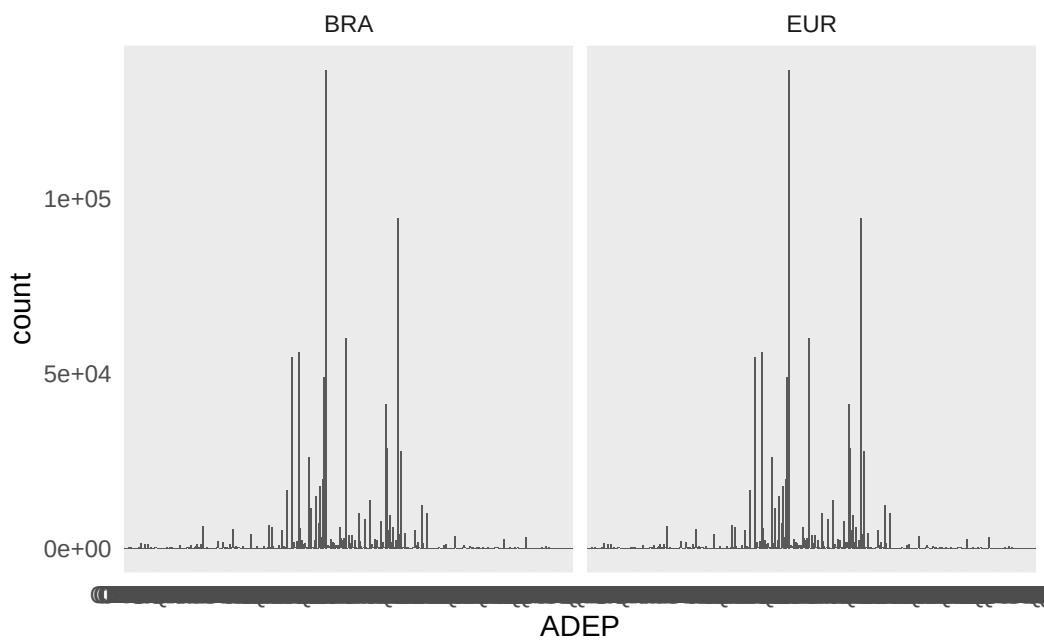
bra_demo <- bra_mvts |> calc_adeq_counts() |> mutate(REGION = "BRA")
eur_demo <- bra_mvts |> calc_adeq_counts() |> mutate(REGION = "EUR")

p1 <- bra_demo |> plot_barchart()
p2 <- eur_demo |> plot_barchart()

p1 / p2
```



```
bind_rows(bra_demo, eur_demo) |> plot_barchart()
```



The maximum departures seen at Brazilian airport account for 136561.